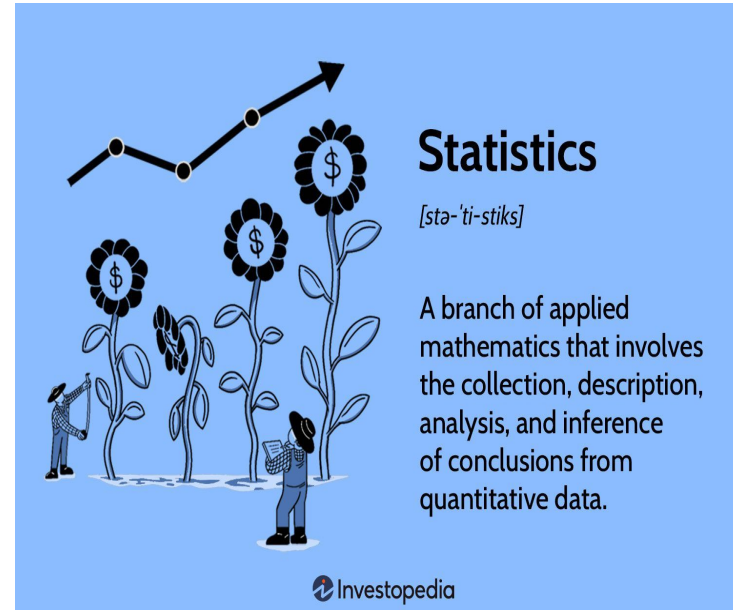


Statistics

What is Statistics?

Statistics is the science of collecting, organizing, analyzing, and interpreting data to make better decisions



Types of Statistics

Descriptive statistics helps us summarize and present data in a meaningful way.

Used For:

- Finding average
- Showing highest & lowest values
- Creating charts and graphs

Example:

“The average score of the class is 75.”

Inferential Statistics

Inferential statistics uses a small data sample to make predictions about a larger population.

Used For:

- Predictions
- Decision making
- Finding trends

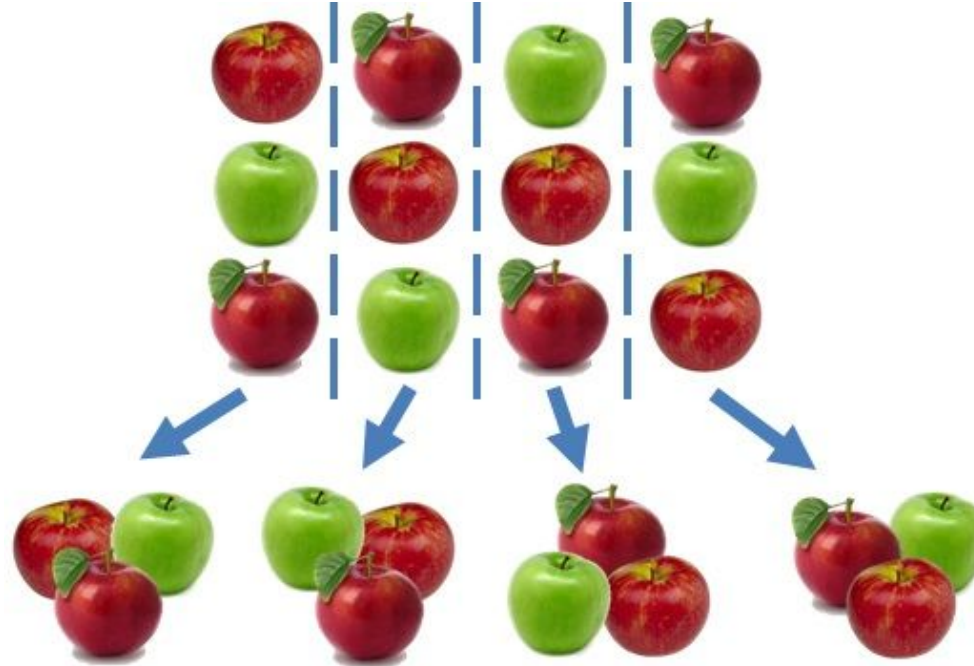
Example:

Survey 100 voters → Predict election result.

Mean (Average)

Mean = Total $\div$ Number of values

use mean to understand the **overall trend**.



Median (Middle Value)

Arrange numbers

Find the middle

Mode (Most Frequent)

2, 4, 4, 6, 8

Mode = 4

👉 **Most sold product**

👉 **Most liked feature**

Percentages (Out of 100)

👉 $(\text{Value} \div \text{Total}) \times 100$

Example:

45 out of 50

= 90%

Used EVERYWHERE:

- ✓ Exams
- ✓ Discounts
- ✓ Growth rate
- ✓ Business reports



Ratios (Comparison Tool)

Ratio compares two quantities.

Example:

Boys = 20
Girls = 10

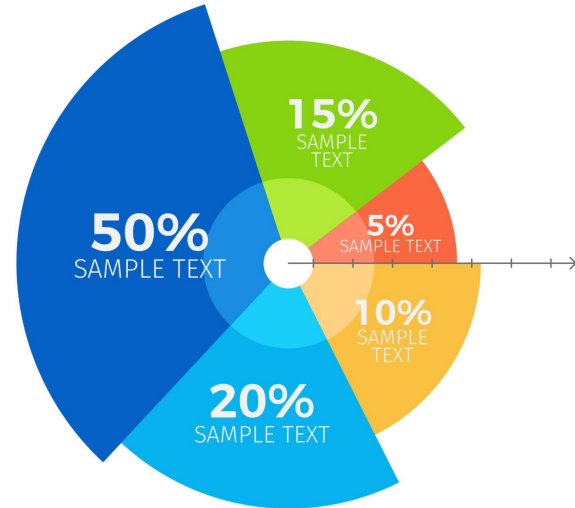
Ratio = 2 : 1

Meaning:

For every 2 boys → 1 girl.

★ Analysts use ratios in:

- Finance
- Population studies
- Business performance



Measures of Central Tendency

👉 **Mean + Median + Mode**

are called:

★ **Measures of Central Tendency**

It just means:

👉 **“Find the center of the data.”**

Measures of Dispersion

Imagine two classes scored the same average marks:

Class A:

👉 70, 72, 71, 69, 70

Class B:

👉 40, 95, 60, 85, 72

Both averages look similar...

But tell me — which class is more consistent?

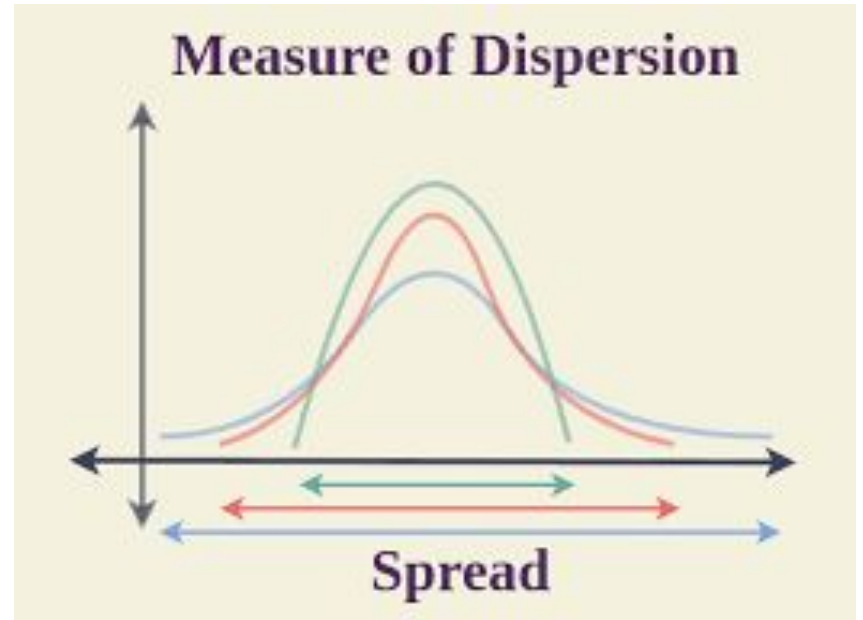
Class A — students perform almost the same.

Class B — performance is all over the place.

👉 This difference in **spread** is called:

Measures of Dispersion

How scattered the data is



Why Dispersion Matters

If you only look at the average...

👉 You see HALF the story.

Dispersion shows the **real behavior of data**.

Used in:

- ✓ Stock market risk
- ✓ Business stability
- ✓ Student performance
- ✓ Machine Learning
- ✓ Medical research

Analysts NEVER trust averages alone.

Range

Range = Highest value – Lowest value

Example:

Marks:

10, 30, 50, 70, 90

Highest = 90

Lowest = 10

👉 Range = 80



What It Tells You:

- ✓ Large range → Data is very spread
- ✓ Small range → Data is tightly packed



Problem With Range:

It only looks at TWO numbers:

- 👉 Smallest
- 👉 Largest

What about the rest?

Ignored ?

So analysts use more advanced tools.

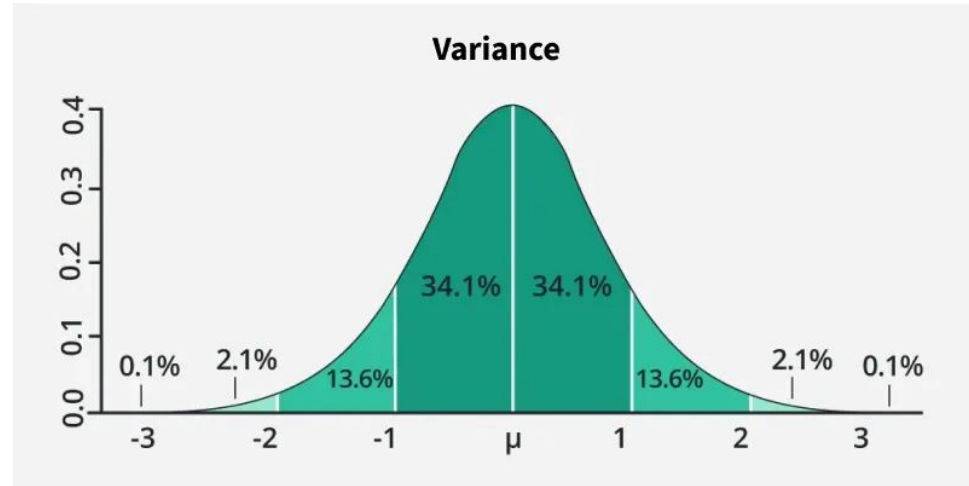
Variance

Measure how far each number is from the **average (mean)**.

Steps analysts imagine:

- 1 Find the mean
- 2 Check how far each value is from it
- 3 Average those distances

That result = **Variance**



What It Means:

- ✓ Small variance → numbers are close to mean
- ✓ Large variance → numbers are far from mean

👉 Helps measure **stability**.

Low Variance:

Daily milk sales = 98, 100, 102

Very predictable business 👍

High Variance:

Sales = 20, 150, 60

Very risky business 😬

Standard Deviation

Average distance of data from the mean.

(It is just the square root of variance — but you don't need to memorize that now.)

Low Standard Deviation:

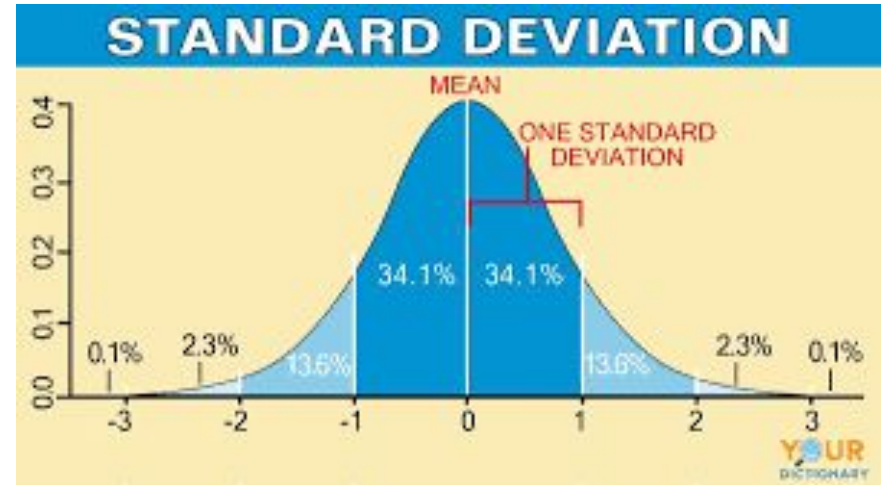
Students score between **75–80**

- ✓ Consistent
- ✓ Reliable

👉 High Standard Deviation:

Scores between **20–95**

- ✗ Unpredictable
- ✗ Risky



Why Experts Love It:

Used in:



Finance → measure investment risk



Machine Learning → understand data patterns



Healthcare → analyze medical data



Sports → player consistency

What is Data Visualization?

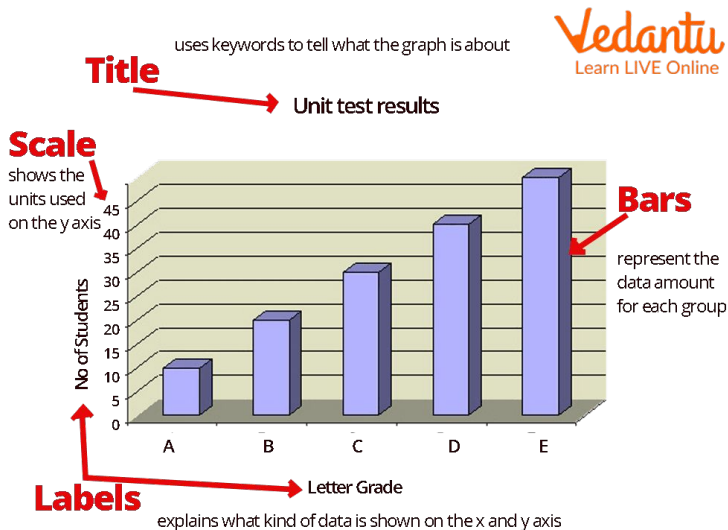
Converting numbers into graphs so the human brain can understand them FAST.

Bar Chart

Uses rectangular bars to compare different categories.

👉 Taller bar = Higher value.

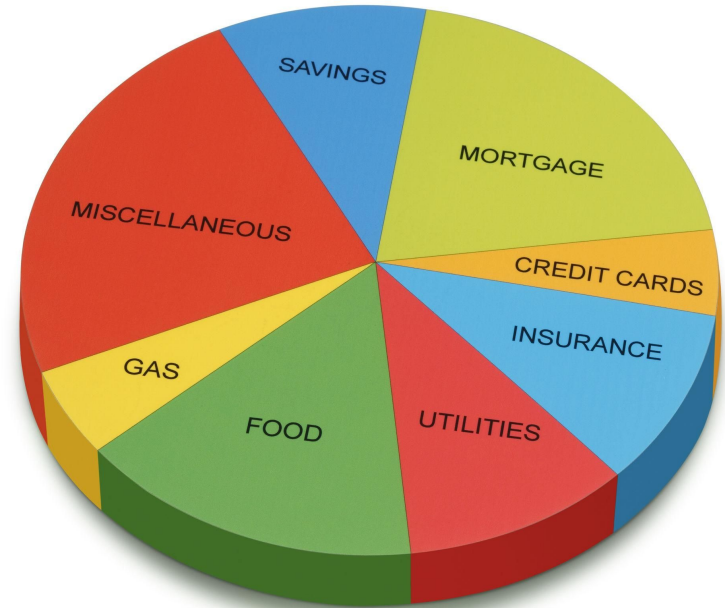
- ✓ Comparing products
- ✓ Student marks
- ✓ Company revenue
- ✓ Survey results



Pie Chart

Each slice shows how much each category contributes.

- 👉 Percentages
- 👉 Market share
- 👉 Budget distribution
- 👉 Population segments



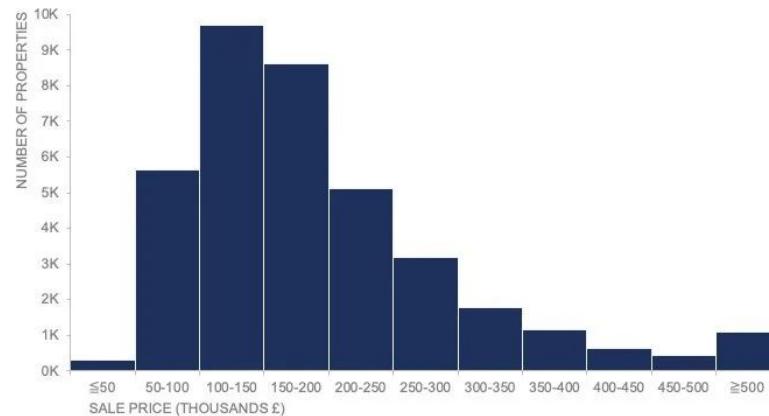
Histogram

Students often confuse this with a bar chart.

👉 **Bar Chart** → compares categories

👉 **Histogram** → shows data distribution

Distribution of property sales: January 2013 to September 2019

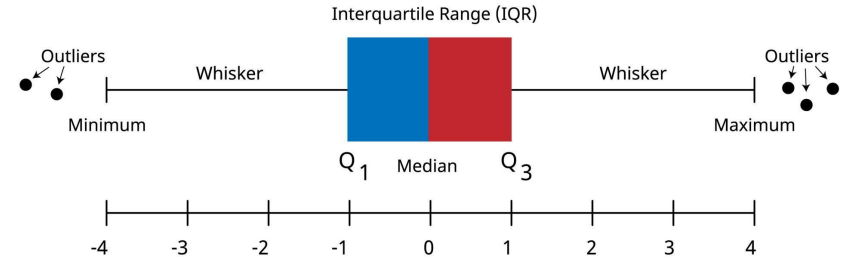


Box Plot

A Box Plot Shows 5 Things:

- ✓ Minimum
- ✓ Maximum
- ✓ Median
- ✓ Quartiles
- ✓ Outliers

Box and Whisker Plots



What is Sampling?

👉 India has **140 crore people**

👉 You want to know: *“Do people like online shopping?”*

We ask a **small group** that represents everyone.

That small group is called a **Sample**

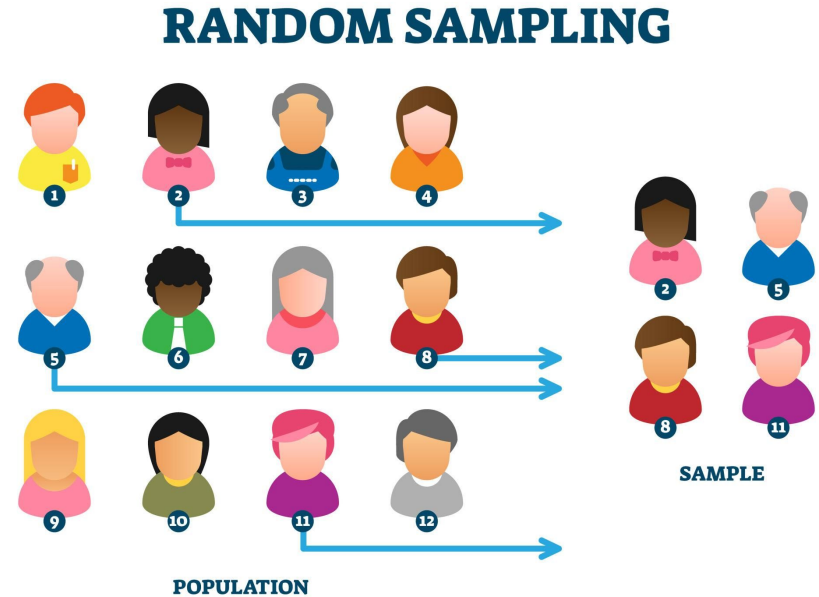
The full population is called **Population**

Simple Random Sampling

Every person has **equal chance** of being selected.

⚠ Limitation:

If population is very large, random sampling can miss important groups.



Stratified Sampling

👉 Population is divided into **groups (strata)**

👉 Sample is taken from **each group**

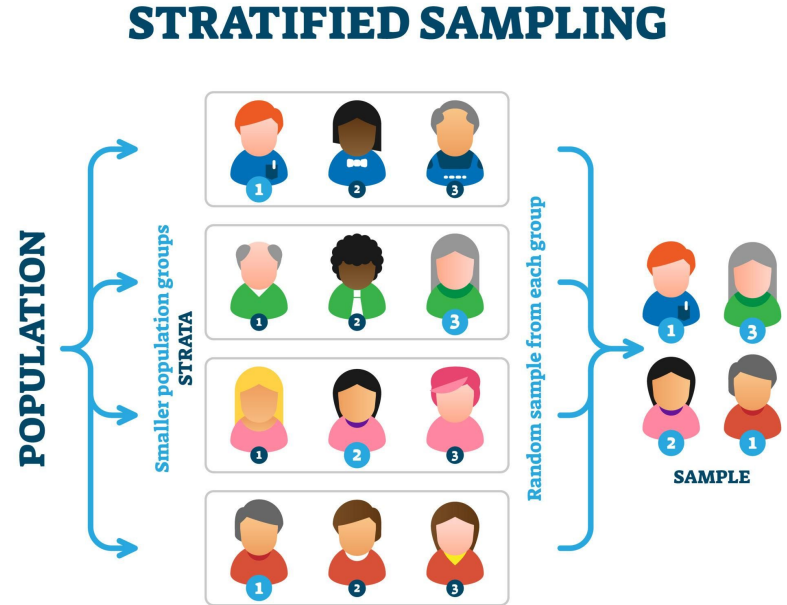
Class has:

- 30 boys
- 20 girls

Teacher wants 10 students.

WE picks:

- 6 boys
- 4 girls

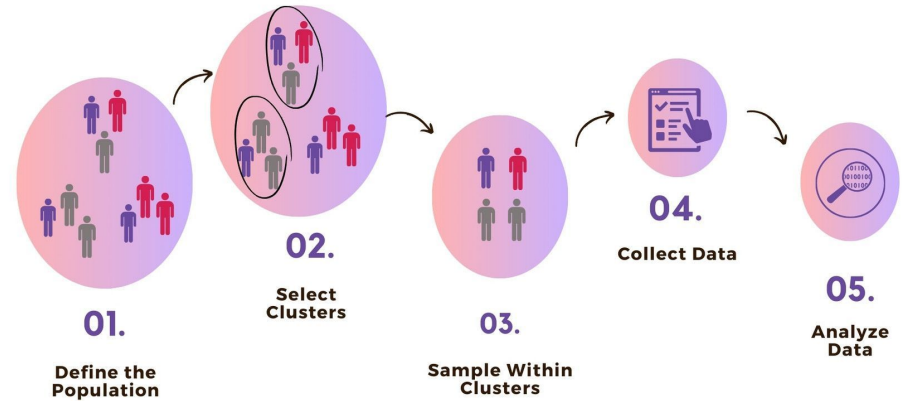


Cluster Sampling

- 👉 Population divided into **clusters**
- 👉 Entire cluster is selected randomly



Steps of Cluster Sampling



Systematic Sampling

Students standing in a line.

Teacher selects:

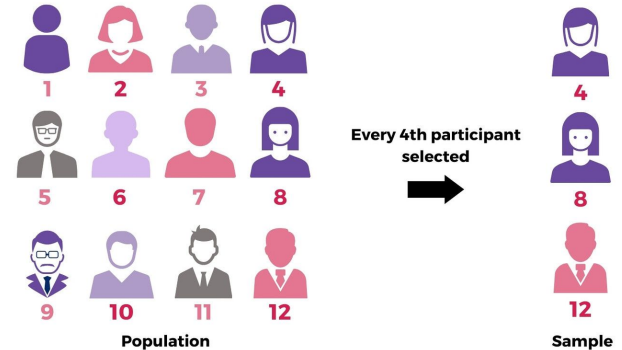
👉 Every 5th student

5, 10, 15, 20...

That's **Systematic Sampling**



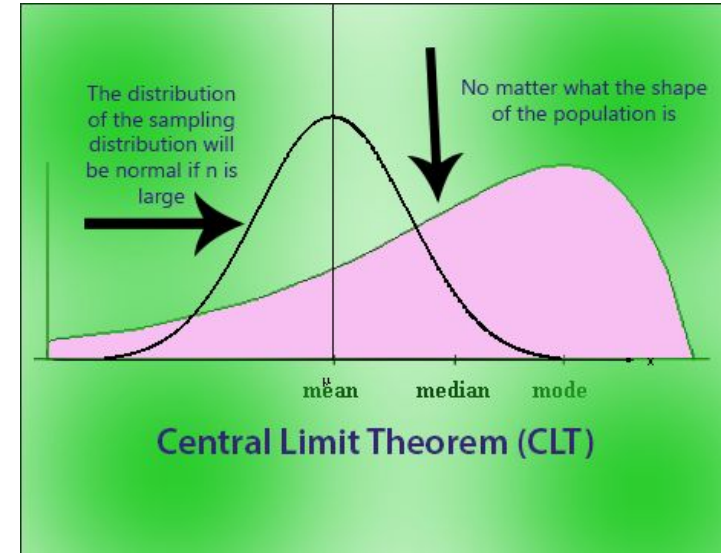
Systematic Random Sampling



What is the Central Limit Theorem

The Central Limit Theorem says:

✓ If you take many random samples from a population, the averages of those samples will form a **NORMAL** (bell-shaped) distribution — even if the original data is messy.



Average height of people in a city.

You take one sample → average = 168 cm

Another sample → 172 cm

Another → 170 cm

Another → 169 cm

Now imagine plotting ALL those sample averages...

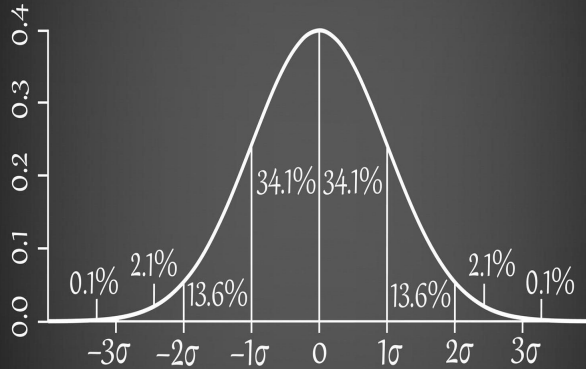
What shape will appear?

👉 **A Bell Curve.**

And that magical result is called:

★ **Central Limit Theorem**

Normal Distribution



- ✓ Most values gather near the center
- ✓ Few values at extremes
- ✓ Perfectly balanced

This is called a **Normal Distribution**.

Probability.

Probability means “chance of something happening.”



Tossing a Coin 🍀

When a coin is tossed, only two outcomes are there: Head or Tail

$$\text{Probability of getting Head} = \frac{1}{2}$$



$$\text{Probability of getting Tail} = \frac{1}{2}$$



There are only **2 possible outcomes**:

- Heads
- Tails

So the probability of getting heads is:

Probability = Favorable Outcomes / Total Outcomes

👉 Here:

Favorable outcome = 1 (Heads)

Total outcomes = 2

✅ **Probability = $1/2 = 0.5 = 50\%$**

0 → Impossible (Example: Sun rising from the west ❌)

0.5 → 50–50 chance

1 → Certain / Guaranteed

Key Terms You MUST Know

👉 **Experiment:** Something you do to see what happens
Example: Tossing a coin.

👉 **Outcome:** Result of the experiment
Example: Heads.

👉 **Event:** One or more outcomes you care about
Example: Getting an even number on a dice (2,4,6).

Independence

Two events are **independent** if one event does **NOT** affect the other.



Conditional Probability

Conditional probability is the probability of an event happening **given that another event has already occurred.**

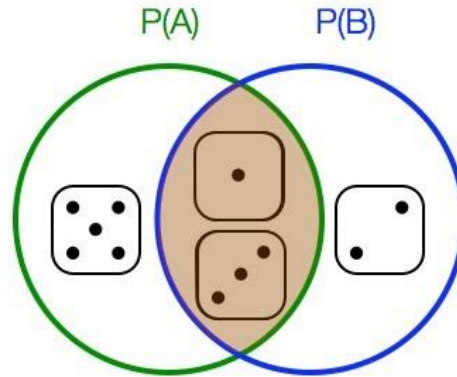


Conditional Probability

What is the Probability of
rolling a dice and it's
value is less than 4

$$P(B | A) = \frac{P(A \cap B)}{P(A)}$$

knowing that the value is
an odd number



Bayes' Theorem

The diagram illustrates Bayes' Theorem with the following components:

- LIKELIHOOD**
The probability of "B" being True, given "A" is True
- PRIOR**
The probability "A" being True. This is the knowledge.
- POSTERIOR**
The probability of "A" being True, given "B" is True
- MARGINALIZATION**
The probability "B" being True.

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Arrows indicate the flow of information: Likelihood and Prior point to the numerator, Posterior points to the left side, and Marginalization points to the denominator.

What is a Probability Distribution?

A **probability distribution** describes how the values of a random variable are distributed.

There are **two main types**:

- **Discrete Distribution**
- **Continuous Distribution**

Discrete Distribution

A **discrete distribution** describes probabilities of **countable values**.

The variable can take **specific, separate values** (usually integers).

Examples:

- Number of students in a class
- Number of cars in a parking lot
- Number of heads when flipping a coin 5 times

Example Distribution:

If you toss a coin 2 times:

Possible outcomes for number of heads:

0, 1, 2

Each has a specific probability.

Common Discrete Distributions:

- Binomial distribution
- Poisson distribution
- Geometric distribution

Continuous Distribution

A **continuous distribution** describes probabilities of **uncountable values**.

The variable can take **any value within an interval** (including decimals).



Examples:

- Height of students
- Weight of a person
- Temperature
- Time taken to complete a task

For example:

Height could be 160.2 cm, 160.23 cm, 160.234 cm... (infinite possibilities)

Example:

$P(160 \text{ cm} < \text{Height} < 170 \text{ cm})$

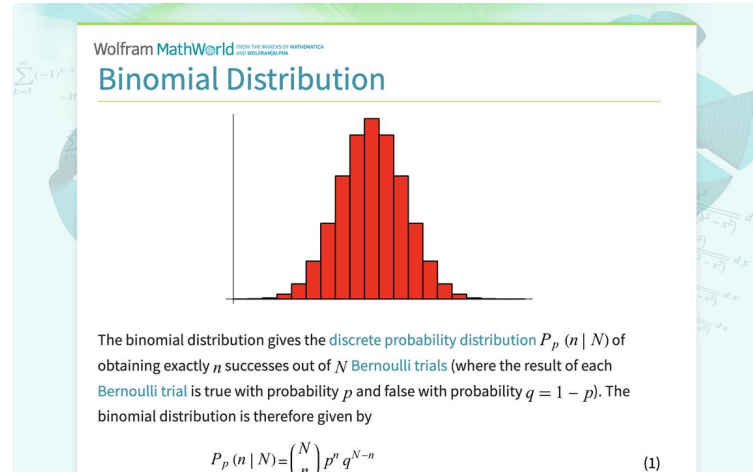
♦ **Common Continuous Distributions:**

- Normal distribution
- Exponential distribution
- Uniform distribution

Binomial distribution

The Binomial distribution models the number of **successes** in a fixed number of independent trials, where:

- Each trial has only **two outcomes** (success/failure)
- Probability of success (**p**) is constant
- Number of trials = **n**



Examples

- Number of heads in 10 coin tosses
- Number of students who pass an exam out of 50
- Number of defective items in a batch

Use Binomial when:

- Fixed number of trials
- Two possible outcomes
- Independent trials

Poisson distribution

The Poisson distribution models the number of times an event occurs in a **fixed interval of time or space**.

It is used for **rare events**.

Examples

- Number of calls received per hour
- Number of accidents at a junction per month
- Number of typing mistakes per page

Use Poisson when:

- Events occur randomly
- Events are independent
- Average rate is known
- Counting occurrences over time/space

Normal distribution

The Normal distribution is a **continuous distribution** that is symmetric and bell-shaped.

It describes many natural phenomena.

Heights of people

Test scores

Measurement errors

When to Use

Use Normal when:

- Data is continuous
- Data is approximately symmetric
- Large sample sizes (Central Limit Theorem effect)